

1 Parameter Setup

This section collects mixture quantities that can be evaluated from the parameter set together with T and $\mathbf{x}$, without solving for density first.

1.1 Base Mixture Quantities

$$\bar{m} = \sum_i x_i m_i \quad (1)$$

$$\overline{MW} = \sum_{j=1}^{N_c} x_j MW_j \quad (2)$$

$$\overline{MW}_{sol} = \sum_{j \in \mathcal{S}} x_j MW_j \quad (3)$$

$$\mathcal{S} = \{j : z_j = 0\}, \quad \mathcal{I} = \{j : z_j \neq 0\} \quad (4)$$

$$x_{sol} = \sum_{j \in \mathcal{S}} x_j \quad (5)$$

1.2 Size and Diameter Rules

This subsection collects the single-component diameter rules that are selected before any ij -mixing is applied.

1.2.1 Segment Diameter

The neutral-segment diameter is the base hard-chain size parameter used in the packing moments and in the pair-contact relations.

$$d_i = \sigma_i \left[1 - 0.12 \exp \left(-3 \frac{\epsilon_i}{kT} \right) \right] \quad (6)$$

$$\left(\frac{\partial d_i}{\partial T} \right)_{\rho, x} \equiv d_{i,T} = \sigma_i \left(3 \frac{\epsilon_i}{kT^2} \right) \left[-0.12 \exp \left(-3 \frac{\epsilon_i}{kT} \right) \right] \quad (7)$$

$$\frac{1}{2} d_i = \left(\frac{d_i d_i}{d_i + d_i} \right) \quad (8)$$

1.2.2 Ion Diameter Rule

The ion-diameter rule selects the effective ionic hard-core size used by the Debye-Huckel and Born electrolyte terms.

$$d_{\text{ion},i} \equiv d_i = \begin{cases} \sigma_i, & \text{sigma rule,} \\ (1 - 0.12) \sigma_i = 0.88 \sigma_i, & \text{constant-factor rule,} \\ \left[1 - 0.12 \exp \left(-3 \frac{\epsilon_i}{kT} \right) \right] \sigma_i, & \text{Barker-Henderson rule,} \end{cases} \quad i \in \mathcal{I} \quad (9)$$

1.3 Pair Mixing Rules

The ordering in this subsection is: pair-diameter and pair-size rules first, pair-energy mixing second, the active ePC-SAFT ionic dispersion override third, and the cross-association mixing rules last.

$$d_{ij} = (d_{ii} + d_{jj}) / 2. \quad (10)$$

$$\sigma_{ij} = \frac{1}{2} (\sigma_i + \sigma_j) \cdot (1 - l_{ij}) \quad (11)$$

$$\epsilon_{ij}^{\text{base}} = \sqrt{\epsilon_i \epsilon_j} (1 - k_{ij}) \quad (12)$$

The active ePC-SAFT implementation then applies a same-sign ionic override to the base rule above. This zero branch does not come from a separate numbered PC-SAFT equation; it comes from the documented native implementation choice in `pair_epsilon_cpp(...)` and is used to remove short-range dispersion for selected ionic pair classes.

$$\epsilon_{ij} = 0 \quad \text{for } z_i z_j > 0, \quad (13)$$

$$\epsilon^{A_i B_j} = \frac{1}{2}(\epsilon^{A_i B_i} + \epsilon^{A_j B_j})(1 - k_{ij}^{\text{hb}}) \quad (14)$$

$$k^{A_i B_j} = \sqrt{k^{A_i B_i} k^{A_j B_j}} \left(\frac{\sqrt{\sigma_i \sigma_j}}{1/2(\sigma_i + \sigma_j)} \right)^3 \quad (15)$$

1.4 Relative Permittivity

The available upstream relative-permittivity rules are grouped first, and the rule-specific supporting expressions then appear in their own subsubsections.

$$\epsilon_r = \begin{cases} \sum_{j=1}^{N_c} \epsilon_{r,j} x_j, & \text{mole-fraction rule,} \\ \frac{\sum_{j=1}^{N_c} \epsilon_{r,j} x_j MW_j}{\overline{MW}}, & \text{mass-fraction rule,} \\ x_{\text{sol}} \epsilon_{r,\text{sol}}^{(w)} + \sum_{j \in \mathcal{I}} \epsilon_{r,j} x_j, & \text{solvent-ion rule,} \\ \frac{\epsilon_{r,\text{solvent,mix}}^{\text{salt-free}}}{1 + 7.01 x_{\text{ion}}}, & \text{ion-suppressed rule.} \end{cases} \quad (16)$$

1.4.1 Mole-Fraction Rule

This branch uses the direct mole-fraction average of the pure-component relative permittivities.

$$\left(\frac{\partial \varepsilon_r}{\partial x_i} \right)_{x_j \neq i} = \varepsilon_{r,i} \quad (17)$$

1.4.2 Mass-Fraction Rule

This branch uses the molar composition together with molar masses to form the mass-weighted relative permittivity.

$$\left(\frac{\partial \varepsilon_r}{\partial x_i} \right)_{x_j \neq i} = \frac{MW_i}{\overline{MW}} (\varepsilon_{r,i} - \varepsilon_r) \quad (18)$$

1.4.3 Solvent-Ion Rule

This branch uses the solvent-only weighted relative permittivity together with the explicit ionic contributions.

$$\left(\frac{\partial \varepsilon_r}{\partial x_i} \right)_{x_j \neq i} = \varepsilon_{r,sol}^{(w)} + x_{sol} \frac{MW_i}{\overline{MW}_{sol}} (\varepsilon_{r,i} - \varepsilon_{r,sol}^{(w)}), \quad i \in \mathcal{S} \quad (19)$$

$$\varepsilon_{r,sol}^{(w)} \equiv \sum_{j \in \mathcal{S}} \varepsilon_{r,j} w_j^{sol} = \frac{\sum_{j \in \mathcal{S}} \varepsilon_{r,j} x_j MW_j}{\sum_{j \in \mathcal{S}} x_j MW_j} = \frac{\sum_{j \in \mathcal{S}} \varepsilon_{r,j} x_j MW_j}{\overline{MW}_{sol}} \quad (20)$$

1.4.4 Ion-Suppressed Empirical Rule

These relations define the optional ion-suppressed empirical relative-permittivity rule and its supporting composition derivatives.

$$\varepsilon_{r,solvent,mix}^{\text{salt-free}} = \sum_{\text{solvent}} w_{\text{solvent}}^{\text{salt-free}} \cdot \varepsilon_{r,solvent} \quad (21)$$

$$\varepsilon_{r,\text{mix}}(\mathbf{x}) = \frac{\varepsilon_{sf}(\mathbf{x})}{1 + 7.01 x_{\text{ion}}(\mathbf{x})}, \quad \varepsilon_{sf} \equiv \varepsilon_{r,\text{solvent},\text{mix}}^{\text{salt-free}} \quad (22)$$

$$\varepsilon_{sf}(\mathbf{x}) = \sum_{s \in \mathcal{S}} w_s^{sf} \varepsilon_{r,s}, \quad w_s^{sf} = \frac{x_s M_s}{\sum_{m \in \mathcal{S}} x_m M_m}, \quad D \equiv \sum_{m \in \mathcal{S}} x_m M_m. \quad (23)$$

$$x_{\text{ion}}(\mathbf{x}) = \sum_{m \in \mathcal{I}} x_m, \quad \left(\frac{\partial x_{\text{ion}}}{\partial x_i} \right)_{T,v,x_{j \neq i}} = \begin{cases} 1, & i \in \mathcal{I}, \\ 0, & i \notin \mathcal{I}. \end{cases} \quad (24)$$

$$\left(\frac{\partial \varepsilon_{sf}}{\partial x_i} \right)_{T,v,x_{j \neq i}} = \sum_{s \in \mathcal{S}} \varepsilon_{r,s} \left(\frac{\partial w_s^{sf}}{\partial x_i} \right)_{T,v,x_{j \neq i}} = \begin{cases} \frac{M_i}{D} (\varepsilon_{r,i} - \varepsilon_{sf}), & i \in \mathcal{S}, \\ 0, & i \notin \mathcal{S}. \end{cases} \quad (25)$$

$$\left(\frac{\partial \varepsilon_{r,\text{mix}}}{\partial x_i} \right)_{T,v,x_{j \neq i}} = \frac{1}{1 + 7.01 x_{\text{ion}}} \left(\frac{\partial \varepsilon_{sf}}{\partial x_i} \right)_{T,v,x_{j \neq i}} - \frac{7.01 \varepsilon_{sf}}{(1 + 7.01 x_{\text{ion}})^2} \left(\frac{\partial x_{\text{ion}}}{\partial x_i} \right)_{T,v,x_{j \neq i}}. \quad (26)$$

$$\left(\frac{\partial \varepsilon_{r,\text{mix}}}{\partial x_i} \right)_{T,v,x_{j \neq i}} = \begin{cases} \frac{1}{1 + 7.01 x_{\text{ion}}} \frac{M_i}{\sum_{m \in \mathcal{S}} x_m M_m} (\varepsilon_{r,i} - \varepsilon_{sf}), & i \in \mathcal{S}, \\ -\frac{7.01 \varepsilon_{sf}}{(1 + 7.01 x_{\text{ion}})^2}, & i \in \mathcal{I}. \end{cases} \quad (27)$$

2 Density

This section collects the density-state relations used downstream. Here ρ is the molar density, ρ_N is the number density in m^{-3} , and $\hat{\rho}$ is the same number density written in $\text{\AA}^{-3}$.

2.1 Packing-Fraction Moments

The hard-chain packing moments are constructed first from the current density state, and their composition and temperature differentials are kept with the same block.

$$\zeta_n = \frac{\pi}{6} \rho \sum_i x_i m_i d_i^n \quad n \in \{0, 1, 2, 3\} \quad (28)$$

$$\zeta_{n,xk} = \left(\frac{\partial \zeta_n}{\partial x_k} \right)_{T, \rho, x_{j \neq k}} = \frac{\pi}{6} \rho m_k (d_k)^n \quad n \in \{0, 1, 2, 3\} \quad (29)$$

$$\zeta_{n,T} = \frac{\partial \zeta_n}{\partial T} = \frac{\pi}{6} \rho \sum_i x_i m_i n d_{i,T} (d_i)^{n-1} \quad n \in \{1, 2, 3\} \quad (30)$$

2.2 Packing Fraction

The third packing moment is the packing fraction used throughout the hard-chain and dispersion relations.

$$\eta = \zeta_3 \quad (31)$$

2.3 Density Conversion

The packing fraction then provides the direct conversion from η to molar density and then to the number-density forms used in the native implementation.

$$\rho = \frac{6}{\pi} \eta \left(\sum_i x_i m_i d_i^3 \right)^{-1} \quad (32)$$

$$\rho_N = N_{AV} \rho, \quad \hat{\rho} = \rho_N \cdot 10^{-30} \quad (33)$$

2.4 Pressure-Density Closure

This final block introduces the pressure-density closure and the residual used for the density root solve.

$$P = ZRT\rho = Zk_B T \rho_N \quad (34)$$

$$Z = 1 + \eta \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial \eta} \right)_{T, x_i} = 1 + \rho \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial \rho} \right)_{T, x_i} \quad (35)$$

In solver form, the density root is the ρ that drives the specified-pressure residual to zero.

$$R_\rho(T, \rho, \mathbf{x}; P^{\text{spec}}) = P^{\text{spec}} - Z(T, \rho, \mathbf{x})RT\rho = 0 \quad (36)$$

3 Contribution Intermediates

3.1 Hard-Chain

This subsection now groups each hard-chain baseline relation with the exact analytical differentials that stem from it.

3.1.1 Hard-Sphere Helmholtz Term $\tilde{a}^{\text{hs}}$

The hard-sphere Helmholtz term is the upstream excluded-volume reference used inside the hard-chain contribution.

$$\tilde{a}^{\text{hs}} = \frac{1}{\zeta_0} \left[\frac{3\zeta_1\zeta_2}{(1-\zeta_3)} + \frac{\zeta_2^3}{\zeta_3(1-\zeta_3)^2} + \left(\frac{\zeta_2^3}{\zeta_3^2} - \zeta_0 \right) \ln(1-\zeta_3) \right] \quad (37)$$

$$\rho \left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial \rho} \right)_{T, x} = \frac{\zeta_3}{(1-\zeta_3)} + \frac{3\zeta_1\zeta_2}{\zeta_0(1-\zeta_3)^2} + \frac{3\zeta_2^3 - \zeta_3\zeta_2^3}{\zeta_0(1-\zeta_3)^3} \quad (38)$$

$$\begin{aligned}
\left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial x_k}\right)_{T, \boldsymbol{\rho}, x_j \neq k} &= -\frac{\zeta_{0,xk}}{\zeta_0} \tilde{a}^{\text{hs}} + \frac{1}{\zeta_0} \left[\frac{3(\zeta_{1,xk}\zeta_2 + \zeta_1\zeta_{2,xk})}{(1-\zeta_3)} + \frac{3\zeta_1\zeta_2\zeta_{3,xk}}{(1-\zeta_3)^2} \right. \\
&\quad + \frac{3\zeta_2^2\zeta_{2,xk}}{\zeta_3(1-\zeta_3)^2} + \frac{\zeta_2^3\zeta_{3,xk}(3\zeta_3-1)}{\zeta_3^2(1-\zeta_3)^3} \\
&\quad + \left(\frac{3\zeta_2^2\zeta_{2,xk}\zeta_3 - 2\zeta_2^3\zeta_{3,xk}}{\zeta_3^3} - \zeta_{0,xk} \right) \ln(1-\zeta_3) \\
&\quad \left. + \left(\zeta_0 - \frac{\zeta_2^3}{\zeta_3^2} \right) \frac{\zeta_{3,xk}}{(1-\zeta_3)} \right]
\end{aligned} \tag{39}$$

$$\begin{aligned}
\left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial T}\right)_{\rho, x_i} &= \frac{1}{\zeta_0} \left[\frac{3(\zeta_{1,T}\zeta_2 + \zeta_1\zeta_{2,T})}{(1-\zeta_3)} + \frac{3\zeta_1\zeta_2\zeta_{3,T}}{(1-\zeta_3)^2} \right. \\
&\quad + \frac{3\zeta_2^2\zeta_{2,T}}{\zeta_3(1-\zeta_3)^2} + \frac{\zeta_2^3\zeta_{3,T}(3\zeta_3-1)}{\zeta_3^2(1-\zeta_3)^3} \\
&\quad \left. + \left(\frac{3\zeta_2^2\zeta_{2,T}\zeta_3 - 2\zeta_2^3\zeta_{3,T}}{\zeta_3^3} \right) \ln(1-\zeta_3) + \left(\zeta_0 - \frac{\zeta_2^3}{\zeta_3^2} \right) \frac{\zeta_{3,T}}{(1-\zeta_3)} \right]
\end{aligned} \tag{40}$$

3.1.2 Hard-Sphere Contact Value g_{ij}^{hs}

The hard-sphere contact value carries the pair-contact geometry used by both the hard-chain Helmholtz expression and the association strength.

$$g_{ij}^{\text{hs}} = \frac{1}{(1-\zeta_3)} + \left(\frac{d_i d_j}{d_i + d_j} \right) \frac{3\zeta_2}{(1-\zeta_3)^2} + \left(\frac{d_i d_j}{d_i + d_j} \right)^2 \frac{2\zeta_2^2}{(1-\zeta_3)^3} \tag{41}$$

$$\begin{aligned}
\rho \left(\frac{\partial g_{ij}^{\text{hs}}}{\partial \rho} \right)_{T,x} &= \frac{\zeta_3}{(1-\zeta_3)^2} + \left(\frac{d_i d_j}{d_i + d_j} \right) \left(\frac{3\zeta_2}{(1-\zeta_3)^2} + \frac{6\zeta_2\zeta_3}{(1-\zeta_3)^3} \right) \\
&\quad + \left(\frac{d_i d_j}{d_i + d_j} \right)^2 \left(\frac{4\zeta_2^2}{(1-\zeta_3)^3} + \frac{6\zeta_2^2\zeta_3}{(1-\zeta_3)^4} \right)
\end{aligned} \tag{42}$$

$$\begin{aligned}
\left(\frac{\partial g_{ij}^{\text{hs}}}{\partial x_k} \right)_{T, \boldsymbol{\rho}, \rho_j \neq k} &= \frac{\zeta_{3,xk}}{(1-\zeta_3)^2} + \left(\frac{d_i d_j}{d_i + d_j} \right) \left(\frac{3\zeta_{2,xk}}{(1-\zeta_3)^2} + \frac{6\zeta_2\zeta_{3,xk}}{(1-\zeta_3)^3} \right) \\
&\quad + \left(\frac{d_i d_j}{d_i + d_j} \right)^2 \left(\frac{4\zeta_2\zeta_{2,xk}}{(1-\zeta_3)^3} + \frac{6\zeta_2^2\zeta_{3,xk}}{(1-\zeta_3)^4} \right)
\end{aligned} \tag{43}$$

$$\begin{aligned}
\left(\frac{\partial g_{ii}^{\text{hs}}}{\partial T}\right)_{\rho,x} &= \frac{\zeta_{3,T}}{(1-\zeta_3)^2} + \left(\frac{1}{2}d_{i,T}\right) \frac{3\zeta_2}{(1-\zeta_3)^2} \\
&+ \left(\frac{1}{2}d_i\right) \left(\frac{3\zeta_{2,T}}{(1-\zeta_3)^2} + \frac{6\zeta_2\zeta_{3,T}}{(1-\zeta_3)^3}\right) + \left(\frac{1}{2}d_id_{i,T}\right) \frac{2\zeta_2^2}{(1-\zeta_3)^3} \\
&+ \left(\frac{1}{2}d_i\right)^2 \left(\frac{4\zeta_2\zeta_{2,T}}{(1-\zeta_3)^3} + \frac{6\zeta_2^2\zeta_{3,T}}{(1-\zeta_3)^4}\right)
\end{aligned} \tag{44}$$

3.2 Dispersion

This subsection groups the dispersion helper polynomials and mixed moments in one place [3].

3.2.1 First-Order Compressibility Prefactor

The C_1 prefactor enters the second-order dispersion term and its density and composition derivatives.

$$C_1 = \left(1 + \bar{m} \frac{8\eta - 2\eta^2}{(1-\eta)^4} + (1 - \bar{m}) \frac{20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4}{[(1-\eta)(2-\eta)]^2}\right) \tag{45}$$

$$C_2 = \frac{\partial C_1}{\partial \eta} = -C_1^2 \left(\bar{m} \frac{-4\eta^2 + 20\eta + 8}{(1-\eta)^5} + (1 - \bar{m}) \frac{2\eta^3 + 12\eta^2 - 48\eta + 40}{[(1-\eta)(2-\eta)]^3}\right) \tag{46}$$

$$\left(\frac{\partial C_1}{\partial x_k}\right)_{T,\rho,x_j \neq k} \equiv C_{1,xk} = C_2\zeta_{3,xk} - C_1^2 \left\{m_k \frac{8\eta - 2\eta^2}{(1-\eta)^4} - m_k \frac{20\eta - 27\eta^2 + 12\eta^3 - 2\eta^4}{[(1-\eta)(2-\eta)]^2}\right\} \tag{47}$$

3.2.2 First Mixed Dispersion Moment

The first mixed dispersion moment carries the $m^2\epsilon\sigma^3$ energy-size average.

$$\overline{m^2\epsilon\sigma^3} = \sum_i \sum_j x_i x_j m_i m_j \left(\frac{\epsilon_{ij}}{kT}\right) \sigma_{ij}^3 \tag{48}$$

$$\left(\frac{\partial \overline{m^2 \epsilon \sigma^3}}{\partial x_k}\right)_{T, \rho, x_j \neq k} \equiv \overline{(m^2 \epsilon \sigma^3)}_{xk} = 2m_k \sum_j x_j m_j \left(\frac{\epsilon_{kj}}{kT}\right) \sigma_{kj}^3 \quad (49)$$

3.2.3 Second Mixed Dispersion Moment

The second mixed dispersion moment carries the $m^2 \epsilon^2 \sigma^3$ energy-size average.

$$\overline{m^2 \epsilon^2 \sigma^3} = \sum_i \sum_j x_i x_j m_i m_j \left(\frac{\epsilon_{ij}}{kT}\right)^2 \sigma_{ij}^3 \quad (50)$$

$$\left(\frac{\partial \overline{m^2 \epsilon^2 \sigma^3}}{\partial x_k}\right)_{T, \rho, x_j \neq k} \equiv \overline{(m^2 \epsilon^2 \sigma^3)}_{xk} = 2m_k \sum_j x_j m_j \left(\frac{\epsilon_{kj}}{kT}\right)^2 \sigma_{kj}^3 \quad (51)$$

3.2.4 First Packing Polynomial

The first packing polynomial I_1 is the η -series used in the leading dispersion term.

$$I_1(\eta, \bar{m}) = \sum_{i=0}^6 a_i(\bar{m}) \eta^i \quad (52)$$

$$\frac{\partial(\eta I_1)}{\partial \eta} = \sum_{j=0}^6 a_j(\bar{m}) (j+1) \eta^j \quad (53)$$

$$\left(\frac{\partial I_1}{\partial x_k}\right)_{T, \rho, x_j \neq k} \equiv I_{1,xk} = \sum_{i=0}^6 [a_i(\bar{m}) i \zeta_{3,xk} \eta^{i-1} + a_{i,xk} \eta^i] \quad (54)$$

3.2.5 Second Packing Polynomial

The second packing polynomial I_2 is the η -series used in the $C_1 I_2$ dispersion term.

$$I_2(\eta, \bar{m}) = \sum_{i=0}^6 b_i(\bar{m}) \eta^i \quad (55)$$

$$\frac{\partial(\eta I_2)}{\partial \eta} = \sum_{i=0}^6 b_j(\bar{m})(j+1)\eta^i \quad (56)$$

$$\left(\frac{\partial I_2}{\partial x_k}\right)_{T,\rho,x_j \neq k} \equiv I_{2,xk} = \sum_{i=0}^6 [b_i(\bar{m})i\zeta_{3,xk}\eta^{i-1} + b_{i,xk}\eta^i] \quad (57)$$

3.2.6 Dispersion Polynomial Coefficients $a_i(\bar{m})$

These coefficients define the $\bar{m}$ -dependent part of the I_1 polynomial.

$$a_i(\bar{m}) = a_{0i} + \frac{\bar{m}-1}{\bar{m}}a_{1i} + \frac{\bar{m}-1}{\bar{m}}\frac{\bar{m}-2}{\bar{m}}a_{2i} \quad (58)$$

$$\left(\frac{\partial a_i}{\partial x_k}\right)_{T,\rho,x_j \neq k} \equiv a_{i,xk} = \frac{m_k}{\bar{m}^2}a_{1i} + \frac{m_k}{\bar{m}^2}\left(3 - \frac{4}{\bar{m}}\right)a_{2i} \quad (59)$$

3.2.7 Dispersion Polynomial Coefficients $b_i(\bar{m})$

These coefficients define the $\bar{m}$ -dependent part of the I_2 polynomial.

$$b_i(\bar{m}) = b_{0i} + \frac{\bar{m}-1}{\bar{m}}b_{1i} + \frac{\bar{m}-1}{\bar{m}}\frac{\bar{m}-2}{\bar{m}}b_{2i} \quad (60)$$

$$\left(\frac{\partial b_i}{\partial x_k}\right)_{T,\rho,x_j \neq k} \equiv b_{i,xk} = \frac{m_k}{\bar{m}^2}b_{1i} + \frac{m_k}{\bar{m}^2}\left(3 - \frac{4}{\bar{m}}\right)b_{2i} \quad (61)$$

3.3 Association

This subsection collects the association site balances, the association-strength relation, and the local differentials that stem from them.

3.3.1 Association Site Fraction

The site-fraction variable X^{A_i} is the fraction of molecules whose site A_i remains unbonded. It is implicit because the unknown site fractions appear on both sides of the mass-action balance, so the full site-fraction vector is solved iteratively at each trial $(T, \rho, \mathbf{x})$ state.

$$X^{A_i} = \left[1 + \sum_j \sum_{B_j} \rho x_j X^{B_j} \Delta^{A_i B_j} \right]^{-1},$$

with $\sum_{B_j}$ over all sites on molecule j ($A_j, B_j, C_j, \dots$),

and $\sum_j$ over all components.

$$\left(\frac{\partial X^{A_i}}{\partial \rho} \right)_{T,x} = -(X^{A_i})^2 \sum_j \sum_{B_j} x_j \left[X^{B_j} \Delta^{A_i B_j} + \rho \left(\Delta^{A_i B_j} \left(\frac{\partial X^{B_j}}{\partial \rho} \right)_{T,x} + X^{B_j} \left(\frac{\partial \Delta^{A_i B_j}}{\partial \rho} \right)_{T,x} \right) \right] \quad (63)$$

$$\left(\frac{\partial X^{A_i}}{\partial x_k} \right)_{T,v,x_{j \neq k}} = -(X^{A_i})^2 \rho \left[\sum_{B_k} X^{B_k} \Delta^{A_i B_k} + \sum_j x_j \sum_{B_j} \left(\Delta^{A_i B_j} \left(\frac{\partial X^{B_j}}{\partial x_k} \right)_{T,v,x_{j \neq k}} + X^{B_j} \left(\frac{\partial \Delta^{A_i B_j}}{\partial x_k} \right)_{T,v,x_{j \neq k}} \right) \right] \quad (64)$$

3.3.2 Component Site Density

The component density ρ_j is the per-component site-density factor used inside the association site balances.

$$\rho_j = x_j \rho_{\text{mixture}} \quad (65)$$

3.3.3 Association Strength

The association strength $\Delta^{A_i B_j}$ collects the pair-diameter, association-volume, Boltzmann, and hard-sphere contact terms into one site-pair interaction measure.

$$\Delta^{A, B_j} = d_{ij}^3 g_{ij}^{\text{hs}} \left[\exp \left(\frac{\epsilon^{A, B_j}}{kT} \right) - 1 \right] \quad (66)$$

$$\left(\frac{\partial \Delta^{A_i B_j}}{\partial \rho} \right)_{T,x} = d_{ij}^3 \kappa^{A_i B_j} \left[\exp(\epsilon^{A_i B_j} / kT) - 1 \right] \left(\frac{\partial g_{ij}^{\text{hs}}}{\partial \rho} \right)_{T,x} \quad (67)$$

$$\left(\frac{\partial \Delta^{A_i B_j}}{\partial x_k} \right)_{T,v,x_j \neq k} = d_{ij}^3 \kappa^{A_i B_j} \left[\exp \left(\frac{\epsilon^{A_i B_j}}{kT} \right) - 1 \right] \left(\frac{\partial g_{ij}^{\text{hs}}}{\partial x_k} \right)_{T,\rho,x_j \neq k} \quad (68)$$

3.4 Debye Huckel

This subsection groups the Debye-Huckel screening expressions with the density, composition, and temperature differentials that stem from them.

3.4.1 Screening Parameter κ

The screening parameter κ is the inverse Debye length used by the Debye-Huckel contribution.

$$\kappa = \sqrt{\frac{\rho e^2}{k_B T \epsilon_0 \epsilon_r} \sum_j x_j z_j^2} \quad (69)$$

$$\begin{aligned}
\rho \left(\frac{\partial \kappa}{\partial \rho} \right)_{T,x} &= \frac{\rho}{2} \left[\frac{\rho e^2}{k_B T \varepsilon_0 \varepsilon_r} \sum_j x_j z_j^2 \right]^{-1/2} \left[\frac{e^2}{k_B T \varepsilon_0 \varepsilon_r} \sum_j x_j z_j^2 - \frac{\rho e^2}{k_B T \varepsilon_0 \varepsilon_r^2} \left(\frac{\partial \varepsilon_r}{\partial \rho} \right)_{T,x} \sum_j x_j z_j^2 \right] \\
&= \frac{\kappa}{2} \left[1 - \frac{\rho}{\varepsilon_r} \left(\frac{\partial \varepsilon_r}{\partial \rho} \right)_{T,x} \right] \\
&= \frac{\kappa}{2} \quad \text{when} \quad \left(\frac{\partial \varepsilon_r}{\partial \rho} \right)_{T,x} = 0
\end{aligned} \tag{70}$$

$$\begin{aligned}
\left(\frac{\partial \kappa}{\partial x_i} \right)_{T,v,x_j \neq i} &= \frac{1}{2} \left(\frac{\rho_N e^2}{k_B T \varepsilon_0 \varepsilon_r} \sum_j x_j z_j^2 \right)^{-\frac{1}{2}} \left\{ -\frac{\rho_N e^2}{k_B T (\varepsilon_0 \varepsilon_r)^2 \varepsilon_0} \frac{\partial (\varepsilon_r)}{\partial x_i} \sum_j x_j z_j^2 + \frac{\rho_N e^2}{k_B T \varepsilon_0 \varepsilon_r} \alpha_i z_i^2 \right\} \\
&= \left(\frac{\rho_N e^2}{k_B T} \right)^{\frac{1}{2}} \left\{ -\frac{1}{2} (\varepsilon_0 \varepsilon_r)^{-\frac{3}{2}} \varepsilon_0 \frac{\partial (\varepsilon_r)}{\partial x_i} \left[\sum_j x_j z_j^2 \right]^{\frac{1}{2}} \right. \\
&\quad \left. + \frac{1}{2 \sqrt{\varepsilon_0 \varepsilon_r}} \left[\sum_j x_j z_j^2 \right]^{-\frac{1}{2}} \alpha_i z_i^2 \right\}
\end{aligned} \tag{71}$$

3.4.2 Finite-Size Correction χ_i

The finite-size correction χ_i modifies the Debye-Huckel free energy for finite ion size, and σ_i is the local $\kappa \chi_i$ derivative helper used by the density and temperature differentials.

$$\chi_i = \frac{3}{(\kappa d_i)^3} \left[\frac{3}{2} + \ln(1 + \kappa d_i) - 2(1 + \kappa d_i) + \frac{1}{2} (1 + \kappa d_i)^2 \right] \tag{72}$$

$$\sigma_i = \left(\frac{\partial (\kappa \chi_i)}{\partial \kappa} \right)_{T,x} = -2\chi_i + \frac{3}{1 + \kappa a_i} \tag{73}$$

$$\rho \left(\frac{\partial \chi_i}{\partial \rho} \right)_{T,x} = \frac{\sigma_i - \chi_i}{2} \tag{74}$$

$$\begin{aligned}
\left(\frac{\partial \chi_i}{\partial x_i}\right)_{T,v,x_j \neq i} &= -\frac{9}{(\kappa d_i)^4} \left[\frac{3}{2} + \ln(1 + \kappa d_i) - 2(1 + \kappa d_i) + \frac{1}{2} (1 + \kappa d_i)^2 \right] d_i \frac{\partial \kappa}{\partial x_i} \\
&\quad + \frac{3}{(\kappa d_i)^3} \left[\frac{1}{1 + \kappa d_i} - 1 + \kappa d_i \right] d_i \frac{\partial \kappa}{\partial x_i} \\
&= 3 \frac{\partial \kappa}{\partial x_i} \left\{ -\frac{\chi_i}{\kappa} + \frac{d_i}{(\kappa d_i)^3} \left[\frac{1}{1 + \kappa d_i} - 1 + \kappa d_i \right] \right\}
\end{aligned} \tag{75}$$

3.5 Born

This subsection collects the Born-model support and the local differentials that stem from the selected Born-mode inputs.

3.5.1 Born Diameter Rule

The Born-diameter rule selects the reference radius used by the base Born and optional SSM/DS terms.

$$d_i^{\text{Born}} = \begin{cases} d_i, & \text{default Born-diameter rule,} \\ d_i^{\text{Born, fitted}}, & \text{fitted Born-diameter rule,} \end{cases} \quad i \in \mathcal{I} \tag{76}$$

3.5.2 Born Mixing Function

The mixing function f_{mix} is the composition-weighted shell-scaling factor used by the optional SSM/DS geometry.

$$f_{mix} = \sum_k x_k f_k \tag{77}$$

3.5.3 Born Shell Shift

The shell shift Δd_i is the optional diameter offset used by the SSM and DS inverse-diameter terms.

$$\Delta d_i = \frac{(f_{mix} - 1)}{|z_i|} \cdot d_i^{\text{Born}} \quad (78)$$

$$\left(\frac{\partial \Delta d_j}{\partial x_i} \right)_{T, \rho, x_{k \neq i}} = \frac{d_j^{\text{Born}}}{|z_j|} \left(\frac{\partial f_{\text{mix}}}{\partial x_i} \right)_{T, \rho, x_{k \neq i}} = \frac{d_j^{\text{Born}}}{|z_j|} f_i, \quad (79)$$

3.5.4 Base Inverse-Diameter Term

The base inverse-diameter term is the core radius term used by the baseline Born model.

$$D_{i, \text{Born}}^{(\text{bulk})} = \frac{1}{d_i^{\text{Born}}}, \quad (80)$$

3.5.5 Solvation-Shell Inverse-Diameter Term

The SSM term applies the shell-shifted inverse-diameter correction in the bulk medium.

$$D_{i, \text{SSM}}^{(\text{bulk})} = \frac{1}{d_i^{\text{Born}} + \Delta d_i} - \frac{1}{d_i^{\text{Born}}}, \quad (81)$$

3.5.6 Dielectric-Saturation Inverse-Diameter Term

The DS term applies the complementary inverse-diameter correction in the ion medium.

$$D_{i, \text{DS}}^{(\text{ion})} = \frac{1}{d_i^{\text{Born}}} - \frac{1}{d_i^{\text{Born}} + \Delta d_i}. \quad (82)$$

4 Residual Helmholtz Energy

This section keeps the final residual Helmholtz contribution expressions together with the five contribution-level density, composition, and temperature differentials assembled from the upstream setup and intermediate blocks.

$$\tilde{a}^{\text{res}} = \tilde{a}^{\text{hc}} + \tilde{a}^{\text{disp}} + \tilde{a}^{\text{assoc}} + \tilde{a}^{\text{DH}} + \tilde{a}^{\text{Born}} \quad (83)$$

The total temperature derivative is assembled from the five contribution-level temperature differentials documented in this section.

$$\left(\frac{\partial \tilde{a}^{\text{res}}}{\partial T} \right)_{\rho, x_i} = \left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial T} \right)_{\rho, x_i} + \left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial T} \right)_{\rho, x_i} + \left(\frac{\partial \tilde{a}^{\text{assoc}}}{\partial T} \right)_{\rho, x_i} + \left(\frac{\partial \tilde{a}^{\text{DH}}}{\partial T} \right)_{\rho, x_i} + \left(\frac{\partial \tilde{a}^{\text{Born}}}{\partial T} \right)_{\rho, x_i} \quad (84)$$

4.1 Hard-Chain Reference Contribution

$$\tilde{a}^{\text{hc}} = \bar{m} \tilde{a}^{\text{hs}} - \sum_i x_i (m_i - 1) \ln g_{ii}^{\text{hs}} (\sigma_{ii}) \quad (85)$$

$$\rho \left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial \rho} \right)_{T, x} = \bar{m} \rho \left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial \rho} \right)_{T, x} - \sum_i x_i (m_i - 1) (g_{ii}^{\text{hs}})^{-1} \rho \frac{\partial g_{ii}^{\text{hs}}}{\partial \rho} \quad (86)$$

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial x_k} \right)_{T, \rho, x_j \neq k} &= m_k \tilde{a}^{\text{hs}} + \bar{m} \left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial x_k} \right)_{T, \rho, x_j \neq k} \\ &\quad - \sum_i x_i (m_i - 1) (g_{ii}^{\text{hs}})^{-1} \left(\frac{\partial g_{ii}^{\text{hs}}}{\partial x_k} \right)_{T, \rho, x_j \neq k} \end{aligned} \quad (87)$$

$$\left(\frac{\partial \tilde{a}^{\text{hc}}}{\partial T}\right)_{\rho, x_i} = \bar{m} \left(\frac{\partial \tilde{a}^{\text{hs}}}{\partial T}\right)_{\rho, x_i} - \sum_i x_i (m_i - 1) (g_{ii}^{\text{hs}})^{-1} \left(\frac{\partial g_{ii}^{\text{hs}}}{\partial T}\right)_{\rho, x_i} \quad (88)$$

4.2 Dispersion Contribution

$$\tilde{a}^{\text{disp}} = -2\pi\rho I_1(\eta, \bar{m}) \overline{m^2 \epsilon \sigma^3} - \pi\rho \bar{m} C_1 I_2(\eta, \bar{m}) \overline{m^2 \epsilon^2 \sigma^3} \quad (89)$$

$$\begin{aligned} \rho \left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial \rho}\right)_{T, x} &= -2\pi\rho \frac{\partial(\eta I_1)}{\partial \eta} \overline{m^2 \epsilon \sigma^3} \\ &\quad - \pi\rho \bar{m} \left[C_1 \frac{\partial(\eta I_2)}{\partial \eta} + C_2 \eta I_2 \right] \overline{m^2 \epsilon^2 \sigma^3} \end{aligned} \quad (90)$$

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial x_k}\right)_{T, \rho, x_j \neq k} &= -2\pi\rho \left[I_{1, xk} \overline{m^2 \epsilon \sigma^3} + I_1 \overline{(m^2 \epsilon \sigma^3)_{xk}} \right] \\ &\quad - \pi\rho \left\{ [m_k C_1 I_2 + \bar{m} C_{1, xk} I_2 + \bar{m} C_1 I_{2, xk}] \overline{m^2 \epsilon^2 \sigma^3} + \bar{m} C_1 I_2 (\bar{m}^2 \epsilon^2 \sigma^3)_{xk} \right\} \end{aligned} \quad (91)$$

$$\left(\frac{\partial \tilde{a}^{\text{disp}}}{\partial T}\right)_{\rho, x_i} = -2\pi\rho \left(I_{1, T} \overline{m^2 \epsilon \sigma^3} + I_1 \overline{m^2 \epsilon \sigma^3}_{, T} \right) - \pi\rho \bar{m} (C_{1, T} I_2 + C_1 I_{2, T}) \overline{m^2 \epsilon^2 \sigma^3} - \pi\rho \bar{m} C_1 I_2 \overline{m^2 \epsilon^2 \sigma^3}_{, T} \quad (92)$$

4.3 Association Contribution

$$\tilde{a}^{\text{assoc}} = \sum_i x_i \sum_{A_i} \left(\ln X^{A_i} - \frac{X^{A_i}}{2} + \frac{1}{2} \right) \quad (93)$$

$$\rho \left(\frac{\partial \tilde{a}^{\text{assoc}}}{\partial \rho}\right)_{T, x} = \rho \sum_i x_i \sum_{A_i} \left(\frac{1}{X^{A_i}} - \frac{1}{2} \right) \left(\frac{\partial X^{A_i}}{\partial \rho} \right)_{T, x} . \quad (94)$$

$$\left(\frac{\partial \tilde{a}^{assoc}}{\partial x_k}\right)_{T,v,x_{i \neq k}} = \sum_{A_k} \left(\ln X^{A_k} - \frac{X^{A_k}}{2} + \frac{1}{2} \right) + \sum_i x_i \sum_{A_i} \left(\frac{1}{X^{A_i}} - \frac{1}{2} \right) \left(\frac{\partial X^{A_i}}{\partial x_k} \right)_{T,v,x_{i \neq k}} \quad (95)$$

$$\left(\frac{\partial \tilde{a}^{assoc}}{\partial T}\right)_{\rho,x_i} = \sum_i x_i \sum_{A_i} \left(\frac{1}{X^{A_i}} - \frac{1}{2} \right) \left(\frac{\partial X^{A_i}}{\partial T} \right)_{\rho,x_i} \quad (96)$$

4.4 Debye and Huckel Contribution

$$\tilde{a}^{DH} = -\frac{\kappa e^2}{12\pi\epsilon_0\epsilon_r k_B T} \sum_i x_i z_i^2 \chi_i \quad (97)$$

$$\rho \left(\frac{\partial \tilde{a}^{DH}}{\partial \rho} \right)_{T,x} = -\frac{e^2}{12\pi\epsilon_0\epsilon_r k_B T} \left[\rho \left(\frac{\partial \kappa}{\partial \rho} \right)_{T,x} \sum_i x_i z_i^2 \chi_i + \kappa \sum_i x_i z_i^2 \rho \left(\frac{\partial \chi_i}{\partial \rho} \right)_{T,x} \right] \quad (98)$$

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{DH}}{\partial x_i} \right)_{T,v,x_{j \neq i}} &= -\frac{e^2}{12\pi\epsilon_0 k_B T} \left[\frac{1}{\epsilon_r} \left(\frac{\partial \kappa}{\partial x_i} \right)_{T,v,x_{j \neq i}} \sum_j x_j z_j^2 \chi_j \right. \\ &\quad - \frac{\kappa}{\epsilon_r^2} \left(\frac{\partial \epsilon_r}{\partial x_i} \right)_{T,v,x_{j \neq i}} \sum_j x_j z_j^2 \chi_j \\ &\quad + \frac{\kappa}{\epsilon_r} \sum_j \chi_j z_j^2 \left(\frac{\partial x_j}{\partial x_i} \right)_{T,v,x_{k \neq i}} \\ &\quad \left. + \frac{\kappa}{\epsilon_r} \sum_j x_j z_j^2 \left(\frac{\partial \chi_j}{\partial x_i} \right)_{T,v,x_{k \neq i}} \right] \quad (99) \end{aligned}$$

$$\left(\frac{\partial \tilde{a}^{DH}}{\partial T} \right)_{\rho,x_i} \approx -\frac{\tilde{a}^{DH}}{T} \quad (100)$$

4.5 Born Contribution

$$\tilde{a}^{Born} = -\frac{e^2}{4\pi\epsilon_0 k_B T} \sum_i x_i z_i^2 \sum_{\delta \in \mathcal{D}} \left(1 - \frac{1}{\epsilon_{r,m_\delta}} \right) D_{i,\delta}^{(m_\delta)}, \quad (101)$$

$$\rho \left(\frac{\partial \tilde{a}^{\text{Born}}}{\partial \rho} \right)_{T,x} = 0 \quad (102)$$

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{\text{Born}}}{\partial x_i} \right)_{T,v_N,x_{j \neq i}} &= -\frac{e^2}{4\pi k_B T \varepsilon_0} \left[\left(1 - \frac{1}{\varepsilon_r} \right) \frac{z_i^2}{d_i^{\text{Born}}} \right. \\ &\quad \left. + \left(\frac{1}{\varepsilon_r^2} \right) \sum_j \frac{x_j z_j^2}{d_j^{\text{Born}}} \left(\frac{\partial \varepsilon_r}{\partial x_i} \right)_{T,v,x_{j \neq i}} \right] \end{aligned} \quad (103)$$

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{\text{Born}}}{\partial x_i} \right)_{T,v,x_{j \neq i}} &= -\frac{e^2}{4\pi \varepsilon_0 k_B T} \left[z_i^2 \left(\left(1 - \frac{1}{\varepsilon_{r,\text{bulk}}} \right) \left[D_{i,\text{Born}}^{(\text{bulk})} + D_{i,\text{SSM}}^{(\text{bulk})} \right] + \left(1 - \frac{1}{\varepsilon_{r,\text{ion}}} \right) D_{i,\text{DS}}^{(\text{ion})} \right) \right. \\ &\quad + \sum_j x_j z_j^2 \left(\frac{1}{\varepsilon_{r,\text{bulk}}^2} \frac{\partial \varepsilon_{r,\text{bulk}}}{\partial x_i} \left[D_{j,\text{Born}}^{(\text{bulk})} + D_{j,\text{SSM}}^{(\text{bulk})} \right] \right. \\ &\quad + \frac{1}{\varepsilon_{r,\text{ion}}^2} \frac{\partial \varepsilon_{r,\text{ion}}}{\partial x_i} D_{j,\text{DS}}^{(\text{ion})} \\ &\quad \left. \left. + \left(\frac{1}{\varepsilon_{r,\text{ion}}} - \frac{1}{\varepsilon_{r,\text{bulk}}} \right) \frac{1}{(d_j^{\text{Born}} + \Delta d_j)^2} \frac{\partial \Delta d_j}{\partial x_i} \right) \right], \end{aligned} \quad (104)$$

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{\text{Born}}}{\partial T} \right)_{\rho,x_i} &= -\frac{\tilde{a}^{\text{Born}}}{T} \\ &\quad - \frac{e^2}{4\pi \varepsilon_0 k_B T} \sum_i x_i z_i^2 \sum_{\delta \in \mathcal{D}} \left[\frac{D_{i,\delta}^{(m_\delta)}}{\varepsilon_{r,m_\delta}^2} \left(\frac{\partial \varepsilon_{r,m_\delta}}{\partial T} \right)_{\rho,x} + \left(1 - \frac{1}{\varepsilon_{r,m_\delta}} \right) \left(\frac{\partial D_{i,\delta}^{(m_\delta)}}{\partial T} \right)_{\rho,x} \right] \end{aligned} \quad (105)$$

The modular Born form is evaluated with the following selectors.

$$\mathcal{D} = \{\text{Born}\} \cup \mathcal{D}_{\text{add}}, \quad \mathcal{D}_{\text{add}} \subseteq \{\text{SSM}, \text{DS}\}, \quad (106)$$

$$m_\delta = \begin{cases} \text{bulk}, & \delta \in \{\text{Born}, \text{SSM}\}, \\ \text{ion}, & \delta = \text{DS}. \end{cases} \quad (107)$$

- $\mathcal{D}$ always contains the base Born term.
- $\mathcal{D}_{\text{add}} = \emptyset$ gives the baseline Born model, $\{\text{SSM}\}$ activates only the solvation-shell correction, $\{\text{DS}\}$ activates only the dielectric-saturation correction, and $\{\text{SSM}, \text{DS}\}$ activates both optional add-ons.
- m_δ selects the relative-permittivity medium used by each active subterm: bulk for $\delta \in \{\text{Born}, \text{SSM}\}$ and ion for $\delta = \text{DS}$.

5 Compressibility Factor

The explicit analytical density differentials now live with their owning setup and contribution-intermediate blocks. This section collects the downstream Z identities using one generic contribution bridge.

$$Z = 1 + \rho \left(\frac{\partial \tilde{a}^{hc}}{\partial \rho} \right)_{T,x} + \rho \left(\frac{\partial \tilde{a}^{disp}}{\partial \rho} \right)_{T,x} + \rho \left(\frac{\partial \tilde{a}^{assoc}}{\partial \rho} \right)_{T,x} + \rho \left(\frac{\partial \tilde{a}^{DH}}{\partial \rho} \right)_{T,x} + \rho \left(\frac{\partial \tilde{a}^{Born}}{\partial \rho} \right)_{T,x} \quad (108)$$

$$Z^\alpha = \rho \left(\frac{\partial \tilde{a}^\alpha}{\partial \rho} \right)_{T,x}, \quad \alpha \in \{hc, disp, assoc, DH, Born\} \quad (109)$$

6 Chemical Potential

This section uses the upstream residual Helmholtz equations together with the colocated density and composition differentials from Sections 4, 1, 2, and 3.

$$\tilde{\mu}_k^{\text{res}} = \frac{\mu_k^{\text{res}}}{kT} = \frac{\hat{\mu}_k^{\text{res}}}{RT} \quad (110)$$

$$\begin{aligned} \tilde{\mu}_k^{\text{res}} = & \tilde{a}^{\text{res}} + (Z - 1) + \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial x_k} \right)_{T, \nu, x_{i \neq k}} \\ & - \sum_{j=1}^N \left[x_j \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial x_j} \right)_{T, \nu, x_{i \neq j}} \right] \end{aligned} \quad (111)$$

$$\tilde{\mu}_k^{\text{res}} = \tilde{\mu}_k^{\text{hc}} + \tilde{\mu}_k^{\text{disp}} + \tilde{\mu}_k^{\text{assoc}} + \tilde{\mu}_k^{\text{DH}} + \tilde{\mu}_k^{\text{Born}} \quad (112)$$

$$\begin{aligned} \tilde{\mu}_k^\alpha = & \tilde{a}^\alpha + Z^\alpha + \left(\frac{\partial \tilde{a}^\alpha}{\partial x_k} \right)_{T, \nu, x_{i \neq k}} \\ & - \sum_{j=1}^N \left[x_j \left(\frac{\partial \tilde{a}^\alpha}{\partial x_j} \right)_{T, \nu, x_{i \neq j}} \right], \quad \alpha \in \{hc, disp, assoc, DH, Born\} \end{aligned} \quad (113)$$

7 Fugacity Coefficient

This section uses the upstream chemical-potential and compressibility identities from Sections 6 and 5.

$$\ln \varphi_k = \tilde{\mu}_k^{\text{res}} - \ln Z = \frac{\mu_k^{\text{res}}}{kT} - \ln Z \quad (114)$$

Using the contribution split for $\tilde{\mu}_k^{\text{res}}$ together with the matching Z^α allocation from Section 5, the total log fugacity coefficient is decomposed as

$$\ln \varphi_k = \ln \varphi_k^{\text{hc}} + \ln \varphi_k^{\text{disp}} + \ln \varphi_k^{\text{assoc}} + \ln \varphi_k^{\text{DH}} + \ln \varphi_k^{\text{Born}} \quad (115)$$

$$\ln \varphi_k^\alpha = \tilde{\mu}_k^\alpha - \frac{Z^\alpha}{Z - 1} \ln Z, \quad \alpha \in \{hc, disp, assoc, DH, Born\} \quad (116)$$

The approximation below is the $Z \rightarrow 1$ limit of Eq. (116). In that limit, $\ln Z/(Z-1) \rightarrow 1$, so the compressibility correction collapses to Z^α . It is only a near-ideal simplification, not the general expression.

$$\ln \varphi_k^\alpha \approx \tilde{\mu}_k^\alpha - Z^\alpha, \quad \alpha \in \{hc, disp, assoc, DH, Born\} \quad (117)$$

8 Activity Coefficient

This section uses the downstream fugacity-coefficient relations from Section 7 together with the selected reference-state definitions.

8.1 Symmetric Reference

$$\gamma_i = \frac{\varphi_i(T, p, \mathbf{x})}{\varphi_{0i}(T, p, x_i = 1)} \quad (118)$$

$$\ln \gamma_i = \ln \varphi_i - \ln \varphi_{0i} \quad (119)$$

8.2 Infinite-Dilution Reference

$$\gamma_i^* = \frac{\varphi_i(T, p, \mathbf{x})}{\varphi_i^\infty(T, p, x_i \rightarrow 0)} \quad (120)$$

$$\ln \gamma_i^* = \ln \varphi_i - \ln \varphi_i^\infty \quad (121)$$

The infinite-dilution reference is the one used for electrolyte solutes and for the contribution-resolved activity-coefficient split below.

$$\ln \gamma_k^* = \ln \gamma_k^{hc,*} + \ln \gamma_k^{disp,*} + \ln \gamma_k^{assoc,*} + \ln \gamma_k^{DH,*} + \ln \gamma_k^{Born,*} \quad (122)$$

$$\gamma_k^{\alpha,*} = \frac{\varphi_k^\alpha}{\varphi_k^{\alpha,\infty}} \quad (123)$$

$$\ln \gamma_k^{\alpha,*} = \ln \varphi_k^\alpha - \ln \varphi_k^{\alpha,\infty} \quad (124)$$

8.3 Mean Ionic Activity Coefficient

$$\gamma_\pm^* = ((\gamma_c^*)^{\nu_c} (\gamma_a^*)^{\nu_a})^{\frac{1}{\nu_c + \nu_a}} \quad (125)$$

$$\ln \gamma_\pm^* = \frac{\nu_c \ln \gamma_c^* + \nu_a \ln \gamma_a^*}{\nu_c + \nu_a} \quad (126)$$

$$\ln \gamma_\pm^{\alpha,*} = \frac{\nu_c \ln \gamma_c^{\alpha,*} + \nu_a \ln \gamma_a^{\alpha,*}}{\nu_c + \nu_a} \quad (127)$$

9 Enthalpy, Entropy, and Gibbs Free Energy

This section uses the upstream residual Helmholtz, compressibility, and colocated temperature-differential identities from Sections 4, 5, and 3.

9.1 Entropy

$$\tilde{h}^{\text{res}} = \frac{\hat{h}^{\text{res}}}{RT} = -T \left(\frac{\partial \tilde{a}^{\text{res}}}{\partial T} \right)_{\rho, x_i} + (Z - 1) \quad (128)$$

9.2 Entropy

$$\tilde{s}^{\text{res}} = \frac{\hat{s}^{\text{res}}(P, T)}{R} = \frac{\hat{s}^{\text{res}}(\nu, T)}{R} + \ln(Z) \quad (129)$$

$$\tilde{s}^{\text{res}} = \frac{\hat{s}^{\text{res}}(P, T)}{R} = -T \left[\left(\frac{\partial \tilde{a}^{\text{res}}}{\partial T} \right)_{\rho, x_i} + \frac{\tilde{a}^{\text{res}}}{T} \right] + \ln(Z) \quad (130)$$

9.3 Gibbs Free Energy

$$\tilde{g}^{\text{res}} = \frac{\hat{g}^{\text{res}}}{RT} = \frac{\hat{h}^{\text{res}}}{RT} - \frac{\hat{s}^{\text{res}}(P, T)}{R} \quad (131)$$

$$\tilde{g}^{\text{res}} = \tilde{a}^{\text{res}} + (Z - 1) - \ln(Z) \quad (132)$$

Gibbs Energy of Solvation

$$\Delta^{\text{solv}, x} G_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0) = RT \ln(\varphi_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0)) \quad (133)$$

Gibbs Energy of Transfer at Infinite Dilution from solvent S1 to solvent S2

$$\Delta^{\text{tr}} G_i^{\infty, \text{S1} \rightarrow \text{S2}}(T, p, x_{j \neq i}, x_i \rightarrow 0) = RT \ln \left(\frac{\varphi_i^{\infty, \text{S2}}}{\varphi_i^{\infty, \text{S1}}} \right) \quad (134)$$

10 Conversions

10.1 Mean Ionic Conversions

This subsection collects the general cation-anion to mean-ionic property conversions used for charged species [1]. The charge-based forms below are the general definitions; for electroneutral salts they reduce to the more familiar stoichiometric mean-ionic expressions used elsewhere in this document.

$$\mu_{\pm} = \frac{\frac{1}{|z_{\text{cat}}|} \mu_{\text{cat}} + \frac{1}{|z_{\text{an}}|} \mu_{\text{an}}}{\frac{1}{|z_{\text{cat}}|} + \frac{1}{|z_{\text{an}}|}} \quad (135)$$

$$f_{\pm} = \left[(f_{\text{cat}})^{1/|z_{\text{cat}}|} (f_{\text{an}})^{1/|z_{\text{an}}|} \right]^{[(1/|z_{\text{cat}}|)+(1/|z_{\text{an}}|)]^{-1}} \quad (136)$$

$$a_{\pm} = \left[(a_{\text{cat}})^{1/|z_{\text{cat}}|} (a_{\text{an}})^{1/|z_{\text{an}}|} \right]^{[(1/|z_{\text{cat}}|)+(1/|z_{\text{an}}|)]^{-1}} \quad (137)$$

$$\varphi_{\pm} = \left[(\varphi_{\text{cat}})^{1/|z_{\text{cat}}|} (\varphi_{\text{an}})^{1/|z_{\text{an}}|} \right]^{[(1/|z_{\text{cat}}|)+(1/|z_{\text{an}}|)]^{-1}} \quad (138)$$

$$\gamma_{\pm} = \left[(\gamma_{\text{cat}})^{1/|z_{\text{cat}}|} (\gamma_{\text{an}})^{1/|z_{\text{an}}|} \right]^{[(1/|z_{\text{cat}}|)+(1/|z_{\text{an}}|)]^{-1}} \quad (139)$$

10.2 Mole-Fraction and Molality Basis

This subsection collects the basis changes between mole-fraction and molality conventions for mean-ionic electrolyte quantities [2].

$$\gamma_{\pm}^{*,m} = \frac{\gamma_{\pm}^{*,x}}{1 + M_{\text{solvent}} \cdot \tilde{m}_{\text{solute}} \cdot \sum_i \nu_{i,\text{ion}}} \quad (140)$$

$$\ln \gamma_{\pm}^{*,m} = \ln \gamma_{\pm}^{*,x} - \ln \left(1 + M_{\text{solvent}} \cdot \tilde{m}_{\text{solute}} \cdot \sum_i \nu_{i,\text{ion}} \right) \quad (141)$$

11 Supplemental

11.1 Legacy / Paper-Direct Expressions

These expressions are retained for traceability to the original literature forms and older derivations. They are useful for comparison to the published equations, but they are not the preferred active documentation path when the main sections above now use reorganized or explicitly derived forms.

11.1.1 Debye-Huckel Density Differential

$$\rho \left(\frac{\partial \tilde{a}^{DH}}{\partial \rho} \right)_{T,x} = -\frac{\kappa e^2}{24\pi k T \epsilon} \sum_i x_i z_i^2 \sigma_i \quad (142)$$

11.1.2 Cameretti 2005 Chemical-Potential Forms

$$\tilde{\mu}_i^{DH} = -\frac{q_i^2 \kappa}{24\pi k T \epsilon} \left[2\chi_i + \frac{\sum_j x_j q_j^2 \sigma_k}{\sum_j x_j q_j^2} \right] \quad (143)$$

$$\sigma_k = \left(\frac{\partial (\kappa \chi_k)}{\partial \kappa} \right)_{T, N} = -2\chi_k + \frac{3}{1 + \kappa d_k} \quad (144)$$

11.2 Bjerrum Treatment

11.2.1 State and Residual Helmholtz Relations

$$R_i = \begin{cases} d_{\text{ion},i}, & d_{\text{ion},i} > l_B, \\ l_B, & d_{\text{ion},i} < l_B. \end{cases} \quad (145)$$

$$l_B = \frac{|z_i z_j| e^2}{8\pi \epsilon_0 \epsilon_r k_B T} \quad (146)$$

$$\tilde{a}^{DH} = -\frac{\kappa e^2}{12\pi \epsilon_0 \epsilon_r k_B T} \sum_i \alpha_i x_i z_i^2 \chi_i \quad (147)$$

$$\kappa = \sqrt{\frac{\rho e^2}{k_B T \epsilon_0 \epsilon_r} \sum_j \alpha_j x_j z_j^2} \quad (148)$$

$$\chi_i = \frac{3}{(\kappa R_i)^3} \left[\frac{3}{2} + \ln(1 + \kappa R_i) - 2(1 + \kappa R_i) + \frac{1}{2} (1 + \kappa R_i)^2 \right] \quad (149)$$

11.2.2 Density Differential

$$\rho \left(\frac{\partial \tilde{a}^{DH}}{\partial \rho} \right)_{T,x} = -\frac{\kappa e^2}{24\pi k T \epsilon} \sum_i \alpha_i x_i z_i^2 \sigma_i \quad (150)$$

$$\sigma_i = \left(\frac{\partial(\kappa\chi_i)}{\partial\kappa} \right)_{T,x} = -2\chi_i + \frac{3}{1 + \kappa R_i} \quad (151)$$

11.2.3 Composition Differential

$$\begin{aligned} \left(\frac{\partial \tilde{a}^{DH}}{\partial x_i} \right)_{T,v,x_{j \neq i}} &= -\frac{e^2}{12\pi\epsilon_0 k_B T} \left[\frac{1}{\epsilon_r} \left(\frac{\partial \kappa}{\partial x_i} \right)_{T,v,x_{j \neq i}} \sum_j \alpha_j x_j z_j^2 \chi_j \right. \\ &\quad - \frac{\kappa}{\epsilon_r^2} \left(\frac{\partial \epsilon_r}{\partial x_i} \right)_{T,v,x_{j \neq i}} \sum_j \alpha_j x_j z_j^2 \chi_j \\ &\quad + \frac{\kappa}{\epsilon_r} \alpha_i z_i^2 \chi_i \\ &\quad \left. + \frac{\kappa}{\epsilon_r} \sum_j \alpha_j x_j z_j^2 \left(\frac{\partial \chi_j}{\partial x_i} \right)_{T,v,x_{k \neq i}} \right] \end{aligned} \quad (152)$$

$$\begin{aligned} \left(\frac{\partial \kappa}{\partial x_i} \right)_{T,v,x_{j \neq i}} &= \frac{1}{2} \left(\frac{\rho_N e^2}{k_B T \epsilon_0 \epsilon_r} \sum_j \alpha_j x_j z_j^2 \right)^{-\frac{1}{2}} \\ &\quad \left\{ -\frac{\rho_N e^2}{k_B T (\epsilon_0 \epsilon_r)^2} \epsilon_0 \frac{\partial(\epsilon_r)}{\partial x_i} \sum_j \alpha_j x_j z_j^2 + \frac{\rho_N e^2}{k_B T \epsilon_0 \epsilon_r} \alpha_i z_i^2 \right\} \\ &= \left(\frac{\rho_N e^2}{k_B T} \right)^{\frac{1}{2}} \left\{ -\frac{1}{2} (\epsilon_0 \epsilon_r)^{-\frac{3}{2}} \epsilon_0 \frac{\partial(\epsilon_r)}{\partial x_i} \left[\sum_j \alpha_j x_j z_j^2 \right]^{\frac{1}{2}} \right. \\ &\quad \left. + \frac{1}{2\sqrt{\epsilon_0 \epsilon_r}} \left[\sum_j \alpha_j x_j z_j^2 \right]^{-\frac{1}{2}} \alpha_i z_i^2 \right\} \end{aligned} \quad (153)$$

$$\begin{aligned} \left(\frac{\partial \chi_i}{\partial x_i} \right)_{T,v,x_{j \neq i}} &= -\frac{9}{(\kappa R_i)^4} \left[\frac{3}{2} + \ln(1 + \kappa R_i) - 2(1 + \kappa R_i) \right. \\ &\quad \left. + \frac{1}{2} (1 + \kappa R_i)^2 \right] R_i \frac{\partial \kappa}{\partial x_i} + \frac{3}{(\kappa R_i)^3} \left[\frac{1}{1 + \kappa R_i} - 1 + \kappa R_i \right] \\ R_i \frac{\partial \kappa}{\partial x_i} &= 3 \frac{\partial \kappa}{\partial x_i} \left\{ -\frac{\chi_i}{\kappa} + \frac{R_i}{(\kappa R_i)^3} \left[\frac{1}{1 + \kappa R_i} - 1 + \kappa R_i \right] \right\} \end{aligned} \quad (154)$$

11.2.4 Ion-Pairing Relations

$$\alpha = \frac{-1 + \sqrt{1 + 4x_{\pm}K_{ip}\frac{(\gamma_{\pm}^*(x_f))^2}{\gamma_{ip}^*}}}{2x_{\pm}K_{ip}\frac{(\gamma_{\pm}^*(x_f))^2}{\gamma_{ip}^*}} \quad (155)$$

$$K_{ip}(T) = \frac{(1 - \alpha) \cdot \gamma_{ip}^*}{\alpha^2 \cdot x_{\pm} \cdot (\gamma_{\pm}^*(x_f(\alpha)))^2} = 4\pi\rho_N \int_a^{l_B} \exp\left(\frac{|z_i z_j| e^2}{4\pi\epsilon_0\epsilon_r k_B T} \cdot \frac{1}{r}\right) r^2 dr \quad (156)$$

$$\gamma_{ip}^* \approx 1 \quad (157)$$

$$x_{\pm} = x_f + x_{ip} \quad (158)$$

$$x_{\pm} = \alpha x_f \quad (159)$$

$$x_{\pm} = (1 - \alpha)x_{ip} \quad (160)$$

$$x_{\pm} = (x_c^{\nu c} \cdot x_a^{\nu a})^{\frac{1}{\nu_c + \nu_a}} \quad (161)$$

$$\mu_i^{DH}(x_f) = \left(\frac{\partial A^{DH}}{\partial \rho_i(x_f)}\right)_{T,V,N_{j \neq i}} = -\frac{e^2 z_i^2 \kappa}{24\pi\epsilon_0\epsilon_r} \left[2\chi_i + \frac{\sum_k x_{k,f} z_k^2 \sigma_k}{\sum_k x_{k,f} z_k^2}\right] \quad (162)$$

$$\ln \gamma_i^*(x_{f,i}) = -\frac{e^2 z_i^2 \kappa}{24\pi\epsilon_0\epsilon_r} \left[2\chi_i + \frac{\sum_k x_{k,f} z_k^2 \sigma_k}{\sum_k x_{k,f} z_k^2}\right] \quad (163)$$

$$\gamma_{\pm}^* = (\gamma_c^{*,\nu c} \cdot \gamma_a^{*,\nu a})^{\frac{1}{\nu_c + \nu_a}} \quad (164)$$

References

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